

Supplement to the Sixth Thought Experiment: Does the State Generation Change When the Normalization Reference Is Changed from Mass to a Charge Unit? - A Restricted Renormalization from $G = M = c = 1$ to $G = q_0 = c = 1$, and an Exact-Match Verification of the Initial States of the Five Stored Conditions -

Noriaki Kihara (ORCID 0009-0004-6753-4020)

2026-09-27 / v1.0 / Version DOI 10.5281/zenodo.22985300

Author: Noriaki Kihara

ORCID: 0009-0004-6753-4020

Version DOI: 10.5281/zenodo.22985300

Concept DOI: 10.5281/zenodo.22985299

Version: v1.0

Date: 2026-09-27

Target paper: The Sixth Thought Experiment v1.1, Version DOI: 10.5281/zenodo.22974632, Concept DOI: 10.5281/zenodo.22974631

Keywords: state closure, normalization, charge unit, mass-to-charge ratio, initial state, charged binary, reproducibility, deterministic state map

Abstract

In the sixth thought experiment [6], under the normalization $G = M = c = 1$, the conservative effects of gravity and Coulomb and the dissipative effects of GW quadrupole and EM dipole radiation of a charged quasi-circular binary were internalized into the eight persistent logical states

$$Z_8 = (U, P, E, H, Q, N, C, D)$$

and a single synchronous state map **transition**(**z**). There it was confirmed, by a code audit and by numerical experiments over five conditions, that **transition**(**z**) receives no external physical argument other than the state z , and that the difference of the physical conditions is given only by the N, C, D stored in the initial state.

In this supplement, for the same physical system, only the normalization reference is changed from

$$G = M = c = 1$$

to

$$\boxed{G = q_0 = c = 1}.$$

Here q_0 is the reference charge unit that counts discrete charges, and in this research

$$q_0 = \frac{e}{3}$$

is adopted. No new dynamics, interaction term, state, update rule, readout or approximation is added at all. The only thing changed is the initialization that makes the initial state from the masses and the charges.

Using equal masses

$$m_A = m_B = \frac{M}{2}$$

and

$$q_A = s_A n q_0, \quad q_B = s_B n q_0, \quad s_A, s_B \in \{+1, -1\},$$

the charge-to-mass ratios of the sixth thought experiment,

$$\lambda_A = \frac{q_A}{m_A}, \quad \lambda_B = \frac{q_B}{m_B},$$

become

$$\lambda_A = s_A \frac{2nq_0}{M}, \quad \lambda_B = s_B \frac{2nq_0}{M}.$$

To preserve the $n = 1$ condition $|\lambda| = 0.30$ of the sixth thought experiment, it suffices to take

$$\boxed{\frac{M}{q_0} = \frac{20}{3}}.$$

Using this single mass-to-charge ratio alone, $|\lambda| = 0.30, 0.90, 1.20$ are regenerated for $n = 1, 3, 4$.

This initialization formula was implemented as an independent initialization program and compared with the `micro=0` rows of the raw data `raw_first_macro_microsteps.csv` of the five conditions stored in the sixth thought experiment. As a result, in all five conditions

$$P, N, C, D, Q$$

and the stored diagnostic columns including the initial work registers,

`micro`, `q_index`, `P`, `N`, `C`, `D`, `k1p`, `k2p`, `k3p`, `k4p`, `dP`, `Q`

matched exactly as IEEE-754 values after reading the CSV.

In the sixth thought experiment it has already been audited that the next state is determined from the present state alone by

$$z_{k+1} = T(z_k).$$

Therefore, at the point where the same initial state z_0 has been generated under the present new normalization, the whole subsequent state sequence is also identical to the existing experiment. Re-running the interaction runs therefore gives no new independent verification, and is not carried out in this supplement.

The conclusion of this supplement is restricted. For the five conditions of the sixth thought experiment, it was confirmed that even when the normalization is changed from $G = M = c = 1$ to $G = q_0 = c = 1$, the same initial states can be generated by giving $M/q_0 = 20/3$, and therefore the existing state evolution is not changed. This is not a derivation of a new physical law; it restates the degree of freedom that had been absorbed into the mass reference in the sixth thought experiment, explicitly, as a mass-to-charge ratio referred to the charge unit.

1. Purpose of this supplement

The normalization of the sixth thought experiment [6] was

$$G = M = c = 1.$$

In this normalization the total mass M is absorbed into the unit, so the charge condition was given to the initial state through

$$\lambda_A = \frac{q_A}{m_A}, \quad \lambda_B = \frac{q_B}{m_B}.$$

This supplement asks only one thing.

$$\boxed{G = M = c = 1 \quad \longrightarrow \quad G = q_0 = c = 1}$$

When the reference is changed in this way and the mass M is kept explicitly, **can exactly the same initial state as in the sixth thought experiment be generated?**

For this question, no new orbit computation or interaction run is needed, because it has already been audited that the generator of the sixth thought experiment holds no physical information other than the state.

The scope of verification of this supplement is therefore only the following three steps.

1. Confirm how the mass and charge dependence was stored into C, D in the initialization code of the sixth thought experiment.
2. Replace only that initialization part by a representation with $G = q_0 = c = 1$ and variable M .
3. Compare the initial rows generated by the new initializer directly with the stored raw initial rows of the sixth thought experiment.

The update map `transition(z)`, the 11-phase machine, the RK4 work registers, the readout, the numerical integration conditions and the existing raw orbit data are not changed.

2. What is already established in the sixth thought experiment

2.1 State closure

In the sixth thought experiment the persistent logical states were taken to be

$$Z_8 = (U, P, E, H, Q, N, C, D),$$

and a full microstate of 41 real components was constructed, including the RK4 work registers and the 11-phase one-hot phase [6].

What matters is that the sole generating map is

`transition(z)`

and that it receives nothing other than the present state z as an argument.

The generating rule can therefore be written abstractly as

$$\boxed{z_{k+1} = T(z_k)}.$$

2.2 The physical condition is stored in the initial state

In the sixth thought experiment the charge dependence was organized as

$$C = 1 - \lambda_A \lambda_B,$$

$$D = (\lambda_A - \lambda_B)^2,$$

where

$$\lambda_A = \frac{q_A}{m_A}, \quad \lambda_B = \frac{q_B}{m_B}.$$

In the implementation, the difference between the five conditions is only the C, D stored in the initial state, and the same `transition(z)` was used for all conditions [6].

In this supplement, therefore, it suffices to regenerate C, D as the same values.

3. Change of the normalization

3.1 The old normalization

The sixth thought experiment adopted

$$G = M = c = 1.$$

Here M is the total mass of the binary. Under the equal-mass condition,

$$m_A = m_B = \frac{M}{2} = \frac{1}{2}.$$

The charge-to-mass ratios used in the five conditions are

$$(\lambda_A, \lambda_B) = (+0.30, -0.30), (+0.30, +0.30), (+0.90, -0.90), (+0.90, +0.90), (+1.20, -1.20).$$

3.2 The new normalization

In this supplement,

$$\boxed{G = q_0 = c = 1}.$$

q_0 is the reference charge unit,

$$q_0 = \frac{e}{3}.$$

The charges are expressed as integer multiples,

$$q_A = s_A n q_0, \quad q_B = s_B n q_0,$$

where

$$s_A, s_B \in \{+1, -1\}, \quad n \in \mathbb{N}.$$

On the other hand, the mass is not absorbed into the unit:

$$m_A = m_B = \frac{M}{2}.$$

Then

$$\lambda_A = \frac{s_A n q_0}{M/2} = s_A \frac{2nq_0}{M},$$

$$\lambda_B = s_B \frac{2nq_0}{M}.$$

The continuous degree of freedom remaining after the change of normalization is therefore

$$\boxed{\frac{M}{q_0}}.$$

4. The mass-to-charge ratio that regenerates the old initial conditions

In the $n = 1$ condition of the sixth thought experiment,

$$|\lambda| = 0.30 = \frac{3}{10}.$$

In the new normalization,

$$|\lambda| = \frac{2q_0}{M}.$$

Hence

$$\frac{2q_0}{M} = \frac{3}{10},$$

that is,

$$\boxed{\frac{M}{q_0} = \frac{20}{3}}.$$

In units with $q_0 = 1$,

$$\boxed{M = \frac{20}{3}},$$

$$\boxed{m_A = m_B = \frac{10}{3}}.$$

Fixing this single value, for a general integer n

$$|\lambda_n| = \frac{2n}{20/3} = \frac{3n}{10}.$$

Therefore

$$n = 1 \Rightarrow |\lambda| = 0.30,$$

$$n = 3 \Rightarrow |\lambda| = 0.90,$$

$$n = 4 \Rightarrow |\lambda| = 1.20.$$

The five conditions of the sixth thought experiment are regenerated as they are from a single M/q_0 .

5. Regeneration of the initial states C, D

Expressing equal and opposite signs by s_A, s_B ,

$$\lambda_A = s_A \frac{3n}{10}, \quad \lambda_B = s_B \frac{3n}{10}.$$

Therefore

$$C = 1 - \lambda_A \lambda_B = 1 - s_A s_B \frac{9n^2}{100},$$

$$D = (\lambda_A - \lambda_B)^2 = \frac{9n^2}{100} (s_A - s_B)^2.$$

For opposite signs $s_A = -s_B$,

$$C = 1 + \frac{9n^2}{100},$$

$$D = \frac{36n^2}{100}.$$

For equal signs $s_A = s_B$,

$$C = 1 - \frac{9n^2}{100},$$

$$D = 0.$$

Hence the five conditions are

case	n	(s_A, s_B)	(λ_A, λ_B)	C	D
n1_attractive	1	(+, -)	(+0.30, -0.30)	1.09	0.36
n1_repulsive	1	(+, +)	(+0.30, +0.30)	0.91	0.00
n3_attractive	3	(+, -)	(+0.90, -0.90)	1.81	3.24
n3_repulsive	3	(+, +)	(+0.90, +0.90)	0.19	0.00
n4_attractive	4	(+, -)	(+1.20, -1.20)	2.44	5.76

which coincide with the initial values of the sixth thought experiment.

6. Code verification

6.1 Scope of the change

In the existing code of the sixth thought experiment, the case table gave $\lambda_A, \lambda_B, C, D$ as in

`("nl_attractive", 0.30, -0.30, 1.09, 0.36)`

and the final C, D were passed to `init_state()`.

In this supplement `transition(z)` is not changed. A new, independent initialization module

`paper6_init_G_q0_c1.py`

was created, which takes as input

$$(q_0, M, n, s_A, s_B)$$

generates

$$\lambda_A, \lambda_B, C, D,$$

and from those values makes the same initial diagnostic row as the sixth thought experiment.

6.2 Test stub

A test stub that generates the five conditions at once,

`test_stub_generate_initial_rows.py`

was created. Its output is

`generated_initial_rows_G_q0_c1.csv`

As the object of comparison, the `micro=0` row was extracted from each

`raw_first_macro_microsteps.csv`

of the five stored cases of the sixth thought experiment. The result of the extraction was fixed as

`original_initial_rows_paper6.csv`

6.3 Method of comparison

With the comparison program

`compare_initial_rows.py`

the following 12 columns were compared.

`micro, q_index, P, N, C, D,`
`k1p, k2p, k3p, k4p, dP, Q`

The comparison was made as an exact match of IEEE-754 floats after reading the CSV as numbers.

Note that $M/q_0 = 20/3$ and $\lambda = 3n/10$ were constructed as rational numbers and converted to float at the end. This is an implementation measure to reproduce decimal notations such as 1.81 as the same IEEE-754 values as the old raw data; it is not a change of the physical model.

7. Verification results

The result was PASS in all five conditions.

case	P	N	C	D	Q	result
n1_attractive	50.0	0.25	1.09	0.36	1.0	PASS
n1_repulsive	50.0	0.25	0.91	0.00	1.0	PASS
n3_attractive	50.0	0.25	1.81	3.24	1.0	PASS
n3_repulsive	50.0	0.25	0.19	0.00	1.0	PASS
n4_attractive	50.0	0.25	2.44	5.76	1.0	PASS

For all cases,

$$\Delta P = \Delta N = \Delta C = \Delta D = \Delta Q = 0$$

and for the initial work registers as well,

$$\Delta k_{1p} = \Delta k_{2p} = \Delta k_{3p} = \Delta k_{4p} = \Delta dP = 0$$

Furthermore,

$$\Delta \text{micro} = 0, \quad \Delta q_{\text{index}} = 0.$$

The initial rows observed in the stored raw data therefore matched exactly in all 12 columns.

The machine-readable detailed results were saved in

`initial_row_verification_result.json`

8. Why the interaction runs are not re-executed

This is an important point of this supplement.

In the sixth thought experiment it has already been audited that the generating map generates the next state from the state z_k alone, as

$$z_{k+1} = T(z_k),$$

and does not refer to G, M, q_A, q_B , the case name or the like as external arguments [6].

Let the initial state under the old normalization be z_0^{old} and the initial state under the new normalization be z_0^{new} .

By the present verification,

$$z_0^{\text{new}} = z_0^{\text{old}}$$

was confirmed for the initialization part that is affected by the change of normalization.

Since the same map T is used,

$$z_1^{\text{new}} = T(z_0^{\text{new}}) = T(z_0^{\text{old}}) = z_1^{\text{old}}.$$

Likewise, by induction,

$$z_k^{\text{new}} = z_k^{\text{old}} \quad (k = 0, 1, 2, \dots)$$

Therefore, re-executing the interaction runs for the five existing conditions would only regenerate the same deterministic state sequence. That is not a new independent verification.

For this reason, in this supplement **only the verification of the initialization equivalence is carried out, and the orbit runs are not re-executed.**

9. Interpretation

9.1 $M = 1$ gave no additional information to the dynamics of the present five conditions

The sixth thought experiment used $M = 1$ for the normalization. In this supplement the charge unit q_0 was set to 1 and, instead,

$$\frac{M}{q_0}$$

was kept.

As a result, with

$$\frac{M}{q_0} = \frac{20}{3}$$

the same initial states as in the sixth thought experiment were obtained.

Therefore, at least for the five conditions verified here, the choice $M = 1$ itself did not give additional independent information to the state evolution. The necessary information can be re-expressed as a dimensionless mass-to-charge ratio at the time of initialization.

9.2 Discrete charge and continuous mass can be described separately

In the new representation, the charge side is expressed by the integer n through

$$q = nq_0,$$

and the mass side can be kept as

$$M/q_0.$$

In the present five conditions, fixing the single

$$M/q_0 = 20/3$$

and only changing $n = 1, 3, 4$ regenerated the old conditions.

However, this supplement does not derive a new quantization rule from this discreteness. What was confirmed here is only the fact that **the existing conditions of the sixth thought experiment can be rewritten, without loss, into an initialization referred to the charge unit.**

10. What this supplement does not show

This supplement does not claim the following.

- That $q_0 = e/3$ has been derived from this theory.
- That $M/q_0 = 20/3$ is a universal physical constant.
- That the origin of mass quantization or charge quantization has been explained.
- Validity for general charged binaries beyond the range of approximation of the sixth thought experiment.
- That a fundamental unification of gravity and electromagnetism has been derived.
- That a new orbit or a new radiation phenomenon has been discovered.

The object of verification of this supplement is only the **equivalence of the initialization normalization** of the five conditions already executed in the sixth thought experiment.

11. Files for reproduction

The files needed to reproduce this supplement were saved in the folder (the folder names in the repository are in Japanese; transliterated here)

04_supplement_Gq0c1_initialization_equivalence_verification_20260927/

inside the storage folder of the sixth thought experiment.

The main files are as follows.

paper6_init_G_q0_c1.py

The restricted initialization implementation that generates the initial states of the five conditions from $G = q_0 = c = 1$, $M/q_0 = 20/3$.

test_stub_generate_initial_rows.py

The test stub that generates the initial rows of the five conditions.

generated_initial_rows_G_q0_c1.csv

The output of the new initializer.

original_initial_rows_paper6.csv

The object of comparison extracted from the stored raw data of the sixth thought experiment.

compare_initial_rows.py

The program verifying the exact match of the new and old initial rows.

initial_row_verification_result.json

The comparison results of all fields.

initial_row_verification_result_ja.md

A short summary of the verification results in Japanese.

12. Conclusion

The normalization of the sixth thought experiment,

$$G = M = c = 1,$$

was changed to

$$G = q_0 = c = 1,$$

and the mass was kept as an explicit degree of freedom.

For the equal-mass binary and the charge unit q_0 ,

$$\lambda_{A,B} = s_{A,B} \frac{2nq_0}{M},$$

and from the reference value $|\lambda| = 0.30$ of the sixth thought experiment,

$$\boxed{\frac{M}{q_0} = \frac{20}{3}}$$

was obtained.

Initializing the five conditions of $n = 1, 3, 4$ with this single ratio, they matched exactly, in all compared columns, the stored initial raw rows of the sixth thought experiment.

Therefore, within the present scope of verification,

$$\boxed{G = M = c = 1 \quad \text{and} \quad G = q_0 = c = 1, \quad M/q_0 = 20/3}$$

are **two normalization representations that generate the same internal initial state**.

Under the state closure

$$z_{k+1} = T(z_k)$$

already established in the sixth thought experiment, if the initial state is the same then all subsequent states are also the same. For this reason, re-execution of the interaction runs is unnecessary.

By this supplement it was confirmed that moving the degree of freedom that had been absorbed into the mass unit in the sixth thought experiment into the single mass-to-charge ratio

$$\boxed{M/q_0}$$

referred to the charge unit does not affect the generation of the physical states of the five existing conditions.

References

This series

[0] Noriaki Kihara, **Designing a Research Project to Explore the Emergence of Spacetime, Particles, and Interactions from States and Relations: A Framework of Preliminary Design and Feasibility Verification for Foundational-Physics Model Exploration under Uncertainty**, v1.0 (2026-09-20). Concept DOI: 10.5281/zenodo.22851944; Version DOI: 10.5281/zenodo.22851945.

[1] Noriaki Kihara, **The First Thought Experiment: Tracing the Equivalence Principle Down to Two Nameless Degrees of Freedom - Deriving the Sign of the Sum of Squares, the Curvature Radius, and Three Systems (Inertial, Uniformly Accelerated, Rotating) from the Conservation of an Area Readout**, v1.0 (2026-09-20). Concept DOI: 10.5281/zenodo.22857952; Version DOI: 10.5281/zenodo.22857953.

[2] Noriaki Kihara, **The Second Thought Experiment: Deriving the Coulomb-Type Inverse-Square Force from a Product Readout and an Area Clock - Keeping Two Values and a**

Linear Law, Obtaining the Focus, Attraction and Repulsion and Neutrality, and Fixing What Two Values Cannot Write, v1.2 (2026-09-21). Concept DOI: 10.5281/zenodo.22867336; Version DOI: 10.5281/zenodo.22876596.

[3] Noriaki Kihara, **The Third Thought Experiment: Reading Out the State of Two Bodies a, b from the Orbit of Two Celestial Bodies a, b - Obtaining the Mass Ratio, the Relative Distance and the Relative Time from Orbit and Gravitational-Wave Readouts, with Two Values of Undetermined Meaning and the Knowledge $G = c = M = 1$ Alone**, v1.5 (2026-09-23). Concept DOI: 10.5281/zenodo.22909660; Version DOI: 10.5281/zenodo.22909661.

[4] Noriaki Kihara, **The Fourth Thought Experiment: From a Fixed Action S to a State-Dependent Action S_n - Can the Relativistic Gravitational Orbits Computed Externally in the Third Thought Experiment Be Regenerated from the Sequential Interaction $X_{n+1} = S_n X_n$?**, v1.0 (2026-09-23). Concept DOI: 10.5281/zenodo.22919787; Version DOI: 10.5281/zenodo.22919788.

[5] Noriaki Kihara, **The Fifth Thought Experiment: Was There No Hidden State? - A Self-Audit of the Two-State Orbit Generation of the Fourth Thought Experiment, Tracking the State Closure from Five States to the Sixth Persistent State $\mathcal{N} = \nu$** , v2.1 (2026-09-26). Concept DOI: 10.5281/zenodo.22973086; Version DOI: 10.5281/zenodo.22973087.

[6] Noriaki Kihara, **The Sixth Thought Experiment: Can Two Kinds of Long-Range Interaction Be Internalized into a Single Closed State Map? - Reconstructing Gravity, Coulomb, GW-Quadrupole and EM-Dipole Radiation of a Charged Quasi-Circular Binary with Eight Persistent States, and Verifying the Transferability of the Same Transition over Five Conditions** -, v1.1 (2026-09-26). Concept DOI: 10.5281/zenodo.22974631; Version DOI: 10.5281/zenodo.22974632.

External

[7] P. C. Peters and J. Mathews, “Gravitational Radiation from Point Masses in a Keplerian Orbit,” *Physical Review* **131**, 435-440 (1963). DOI: 10.1103/PhysRev.131.435.

[8] P. C. Peters, “Gravitational Radiation and the Motion of Two Point Masses,” *Physical Review* **136**, B1224-B1232 (1964). DOI: 10.1103/PhysRev.136.B1224.

[9] L. Liu, O. Christiansen, Z.-K. Guo, R.-G. Cai, and S. P. Kim, “Gravitational and electromagnetic radiation from binary black holes with electric and magnetic charges: Circular orbits on a cone,” *Physical Review D* **102**, 103520 (2020). DOI: 10.1103/PhysRevD.102.103520.

[10] C. A. Benavides-Gallego and W.-B. Han, “Gravitational Waves and Electromagnetic Radiation from Charged Black Hole Binaries,” *Symmetry* **15**, 537 (2023). DOI: 10.3390/sym15020537.

[11] Z.-H. Zhang, T. Liu, S. Zhang, and Z.-K. Guo, “Post-Newtonian dynamics of charged compact binaries,” *Physical Review D* **114**, 044088 (2026). DOI: 10.1103/4x8q-2kzm.

[12] S. Verma et al., “Post-newtonian dynamics of radiating charges: canonical formulation and binary inspiral laws,” *European Physical Journal C* **86**, 830 (2026). DOI: 10.1140/epjc/s10052-026-16076-2.

Reproducibility note

This supplement does not change the dynamics of the existing sixth thought experiment. The new code used for the verification, the generated initial rows, the initial rows extracted from the old raw data, the comparison code and the comparison results were placed in the same storage folder described in Section 11 of the main text. For the interaction-run data, the existing raw data of the sixth thought experiment are taken as the object of verification as they are, and are not regenerated.